

BPM PROJECT

Optimizing the Request-to-Maintain Process at Ford Service Center

MIS431: selected topics in management information system (1)

SECTION: 76200

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1.Process Context

The Request-to-Maintain process represents the end-to-end workflow followed at a Ford Service Center to handle customer maintenance requests. The process begins when a maintenance request is received and includes key stages such as warranty validation, initial inspection, quotation preparation, parts availability, service booking, and payment handling, before the maintenance service is executed or the request is terminated.

The process requires coordination between multiple internal and external entities, including the Service Coordination Department, Warranty Department, Warehouse, Suppliers, and External Garage, ensuring that customer requests are handled efficiently and accurately.

2.Process Boundaries

Start Point:

The process begins when a customer submits a maintenance work order request to the Service Coordination Department.

End Point:

The process ends either when the maintenance service is successfully completed or when the request is terminated due to warranty issues, quotation rejection, parts unavailability, or failure to complete payment within the required timeframe.

Process Scope

The process includes all activities required to handle a customer maintenance request within the service center, ensuring proper coordination, validation, and execution of the request until completion or cancellation.

Main process:

Request-to-Maintain

Sub-Processes:

The main process consists of the following sub-processes:

1. **Handle Invalid Warranty**

Manages cases where the warranty is invalid or not applicable and allows the customer to proceed with self-payment or terminate the request.

2. **Initial Inspection**

Represents the initial vehicle inspection performed to identify required maintenance and determine service needs.

3. **Prepare Quotation**

Uses inspection results to estimate required resources and prepares a quotation for customer approval.

4. **Parts Availability**

Handles the availability and ordering of required spare parts before proceeding with service activities.

5. **Service Booking**

Coordinates with the external garage to arrange the service appointment and required resources.

6. **Payment Handling**

Ensures payment is completed within the specified timeframe before service execution; otherwise, the request is cancelled.

3.Process Profile

Element	Description
Process Name	Request-to-Maintain Process.
Process Type	Core operational process.
Process Owner	Service Coordination Department.
Customers	Vehicle owners requesting maintenance services.
Stakeholders	Warranty Department, Warehouse Department, External Garage, Certified Spare Parts Suppliers (Resellers).
Inputs	Warranty Department, Warehouse Department, External Garage, Certified Spare Parts Suppliers (Resellers).
Outputs	Confirmed service booking, completed maintenance service, or service cancellation notification.
Key Activities	Warranty validation, initial inspection, quotation preparation, parts availability handling, service booking, payment handling, and maintenance execution
Decision Points	Warranty validity (valid/invalid), customer decision (accept/reject quotation), parts availability (available/unavailable), supplier confirmation (confirmed/not confirmed), and payment status (paid/not paid).
Exception Handling	Invalid or non-applicable warranty, customer rejection of quotation, parts unavailability after supplier attempts, and delayed or missing payment leading to process termination
Time Constraints	Customer response within 3 days for warranty-related decisions, response within 5 days for quotation approval or rejection, and payment completion within 1 week after appointment confirmation.
KPIs	• Cycle Time: Total time from request receipt to service completion or cancellation. • Cycle Time Efficiency (CTE): Ratio of actual processing time to overall cycle time. • Quotation Acceptance Rate: Percentage of quotations accepted by customers. • Parts Availability Rate: Percentage of requests fulfilled without parts-related delays. • On-Time Payment Rate: Percentage of payments completed within the one-week deadline. • Resource Utilization Rate: Percentage of working time actively used by staff to process requests.

Table 1Process profile

4.Stakeholders

Stakeholder	Type	Role
Customer	External	Submits maintenance request and decides whether to approve or reject the quotation; completes payment if proceeding.
Service Coordination Department	Internal	Manages the overall process, prepares quotations, coordinates between departments, and communicates with the customer.
Warranty Department	Internal	Verifies warranty validity and determines whether the request is covered
Warehouse Department	Internal	Checks spare parts availability and inform service Coordination Department if parts are not available.
External Garage	External	Provides maintenance services and assigns technicians to perform the required work.
Certified Spare Parts Supplier (Reseller)	External	Supplies spare parts required for the maintenance service , confirms or rejects orders placed by the Warehouse Department within the specified timeframe.

Table 2 Stakeholders

Assumptions:

1. The maintenance request is assumed to contain complete and accurate information required to prepare the quotation.
2. The Service Coordination Department manages communication between the customer and all involved departments.
3. A quotation is prepared and must be accepted by the customer before proceeding with service-related activities.
4. Warranty validation is completed before service booking; however, the process may continue if the customer agrees to proceed with self-payment.
5. Required spare parts must be available or successfully ordered before proceeding to service booking.
6. Payment must be completed within one week after appointment confirmation; otherwise, the maintenance request is cancelled.

5.Process Discovery:

Data Collection Approach

To understand the current “as-is” maintenance process, we relied on observation, supported by prior experience with periodic vehicle maintenance. This helped us understand the general workflow and interactions between involved parties.

This understanding was further supported by evidence-based sources, including online articles and educational videos, which provided insights into standard service practices and helped validate the sequence of activities and decision points.

Combining these two sources allowed us to build a clear and realistic understanding of the process and represent it accurately in the BPMN model.

As-Is Diagram

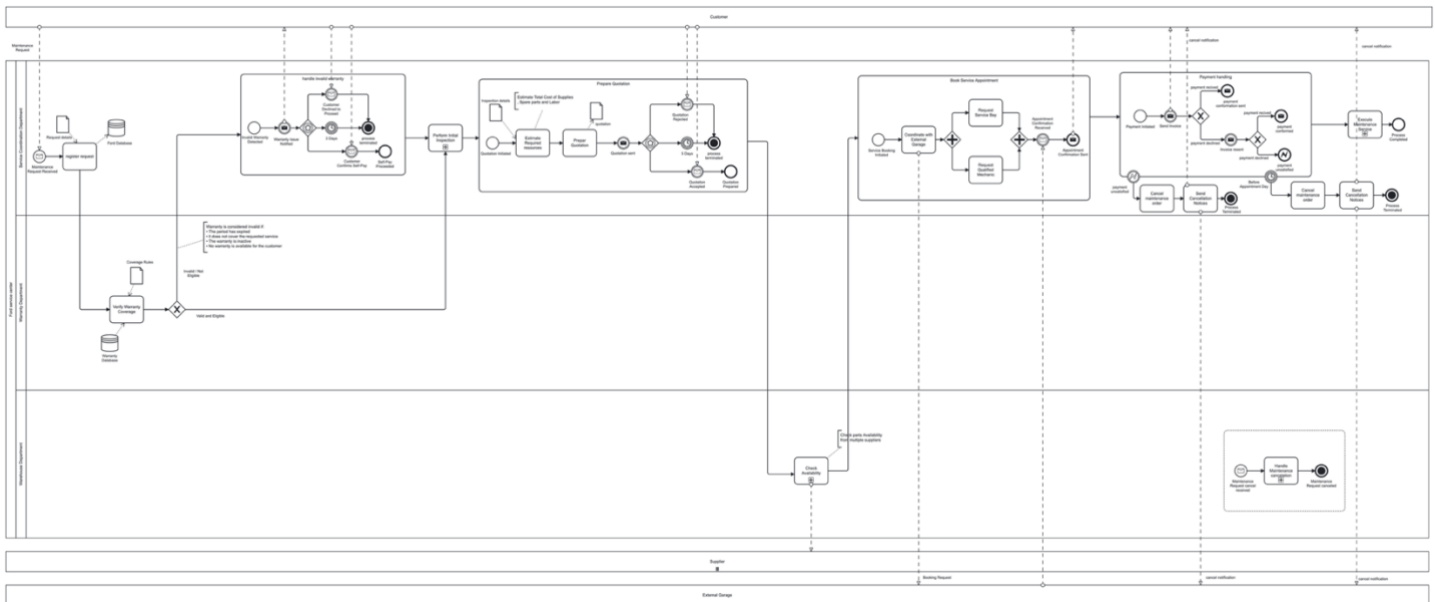


Figure 1 As -Is Diagram

For a clearer and more detailed view of the As-Is diagram, please refer to the full-resolution version available via the link below.

<https://drive.google.com/file/d/1HWXkfNgYgPmKc-7wwEOCzr5eCOUIOVDP/view?usp=drivesdk>

6. Qualitative Analysis

6.1 Value-Added Analysis

	Step	Performer	Classification
1	Receive maintenance request from customer	Customer	NVA
2	Register maintenance request in system	Service Coordination Dept.	BVA
3	Verify warranty coverage against coverage rules & database	Warranty Department	BVA
4	Determine warranty validity (Valid / Invalid)	Warranty Department	BVA
5	Notify customer of warranty issue	Service Coordination Dept.	BVA
6	Wait for customer response (Decline / Self-Pay / 3-Day timeout)	Service Coordination Dept.	NVA
7	Process customer decision (Proceed with self-pay or terminate)	Service Coordination Dept.	VA
8	Conduct initial vehicle inspection	Service Coordination Dept.	VA
9	Estimate required resources (spare parts, supplies, labor)	Service Coordination Dept.	BVA
10	Prepare quotation document	Service Coordination Dept.	VA
11	Send quotation to customer	Service Coordination Dept.	NVA
12	Wait for customer response (Accept / Reject / 5-Day timeout)	Service Coordination Dept.	NVA
13	Process customer decision on quotation	Service Coordination Dept.	VA
14	Check parts availability from multiple suppliers	Warehouse Department	BVA
15	Place parts order request to supplier	Warehouse Department	BVA
16	Coordinate with external garage for scheduling	Service Coordination Dept.	BVA
17	Request service bay allocation	Service Coordination Dept.	BVA
18	Request qualified mechanic assignment	Service Coordination Dept.	BVA
19	Receive appointment confirmation from external garage	Service Coordination Dept.	NVA
20	Send appointment confirmation to customer	Service Coordination Dept.	NVA
21	Send invoice to customer	Service Coordination Dept.	BVA

22	Receive payment from customer	Service Coordination Dept.	VA
23	Process payment confirmation	Service Coordination Dept.	BVA
24	Resend invoice if payment declined (retry)	Service Coordination Dept.	NVA
25	Cancel maintenance order if payment unsatisfied	Service Coordination Dept.	NVA
26	Send cancellation notice to customer & external garage	Service Coordination Dept.	NVA
27	Perform vehicle maintenance/repair service	Service Coordination Dept.	VA
28	Receive cancellation request from customer	Warehouse Department	NVA
29	Handle maintenance cancellation process	Warehouse Department	BVA

Value added analysis 3 Table

6.2 Waste Analysis

Unnecessary Transportation	• The maintenance request is manually escalated from the Service Coordination Department to the Warranty Department for coverage verification, instead of being automatically routed through the system.
Waiting	• Waiting 3 days for the customer to respond after the warranty issue notification before deciding whether to proceed with self-pay or terminate the request. • Waiting 5 days for the customer to accept or reject the quotation. • Waiting for appointment confirmation after submitting separate requests for the service bay and the qualified mechanic. • Waiting for payment after the invoice is issued, with the possibility of timeout leading to cancellation.
Motion	• Manual email communication is used by the department to coordinate with the external garage instead of using an integrated system communication channel.
Inventory	• Maintenance requests remain pending while waiting for quotation approval or customer decisions. • Spare parts availability is confirmed only after quotation acceptance, which may delay subsequent steps if parts are unavailable.
Defect	• Payment failures represent a defect in the process, as the transaction does not complete successfully and requires the invoice to be resent and the payment step to be repeated.
Over-Processing	• Appointment booking requires two separate requests (service bay and qualified mechanic), increasing process complexity unnecessarily. • A full inspection and detailed quotation are prepared before confirming the customer's intention to proceed, which may result in wasted effort if the request is ultimately not accepted. • Warranty verification is performed for all requests, even when the customer may choose to proceed with self-payment, resulting in unnecessary processing in some cases.

Table 4 Waste Analysis

6.3 Stakeholder Analysis

The table below identifies the five stakeholders involved in the Request-to-Maintain process and summarizes their main concerns.

Stakeholder	Key Issues / Concerns
1. Customer	• Long waiting time to receive quotation and appointment confirmation. • Unclear or no updates on service progress. • Risk of cancellation if payment is not completed within one week.
2. Service Coordination Department	• High workload from manually managing all communication with all parties. • Delays caused by waiting for customer responses at multiple steps. • Sending multiple cancellation notices increases operational effort.
3. Warranty Department	• Requests are routed manually instead of through the system. • Must verify coverage even when the customer intends to self-pay. • Incomplete customer data reduces accuracy of warranty decisions.
4. Warehouse Department	• Parts availability is checked only after the customer approves the quotation. • Suppliers are contacted one by one, which increases the waiting time. • Parts unavailability at a late stage can block the entire service booking.
5. External Garage	• Booking requests are received via manual email with no integrated system. • Two separate requests are needed for the service bay and the mechanic. • Late cancellation notices disrupt resource scheduling.
6. Certified Spare Parts Supplier (Reseller)	• Orders are placed sequentially (one supplier at a time), causing delays when the first supplier cannot fulfill the order. • No integrated system for receiving and confirming orders, relying on manual communication.

Table 5 Stakeholders Analysis

6.4 Issue Documentation

Issue Name	Explanation	Assumptions	Qualitative Impact	Quantitative Impact
Long Waiting Times	The process waits up to 3 days for the customer to respond to a warranty issue, and up to 5 days for quotation approval. These delays make up most of the 15.9-day cycle time.	500 requests per year. 30% have invalid warranty (3-day wait). - All requests require quotation approval (5-day wait). - Staff cost = 50 SAR/hour.	- Low customer satisfaction. - High number of pending cases (WIP). - Process efficiency drops to only 10.3%.	Avg wait/case: $(3 \times 0.30) + 5 = 5.9$ days Annual wasted days: $500 \times 5.9 = 2,950$ days Annual cost: $2,950 \times 8\text{h} \times 50\text{ SAR} = 1,180,000\text{ SAR/year}$
Late Parts Availability Check	Spare parts availability is only checked after the customer approves the quotation. If parts are unavailable, the process is delayed at a late stage when the customer already expects the service to proceed.	500 requests per year. 15% face parts unavailability. - Each case is delayed by 2 working days. - Staff cost = 50 SAR/hour.	- Customer frustration from unexpected delays. - Service booking cannot proceed until resolved. - Last-minute orders strain supplier relationships.	Affected cases: $500 \times 15\% = 75$ cases/year Wasted time: $75 \times 2\text{ days} \times 8\text{h} = 1,200\text{h}$ Annual cost: $1,200 \times 50\text{ SAR} = 60,000\text{ SAR/year}$
Split Service Booking Requests	Scheduling with the External Garage requires two separate requests — one for the service bay and one for the mechanic. This adds unnecessary steps and increases the risk of scheduling conflicts.	500 bookings per year. - Each split request wastes ~20 min. 10% of cases result in a scheduling conflict. - Each conflict takes ~1h to resolve. - Staff cost = 50 SAR/hour.	- Unnecessary complexity in the booking step. - Higher chance of misaligned garage scheduling. - Extra coordination effort for Service Coordination.	Wasted time (split requests): $500 \times 20\text{ min} = 10,000\text{ min}$ $= 167\text{h} \times 50 = 8,350\text{ SAR/year}$ Conflict rework: $500 \times 10\% \times 1\text{h} \times 50\text{ SAR} = 2,500\text{ SAR/year}$ Total: ~10,850 SAR/year

Table 6 Issue Documentation

6.5 Root cause analysis

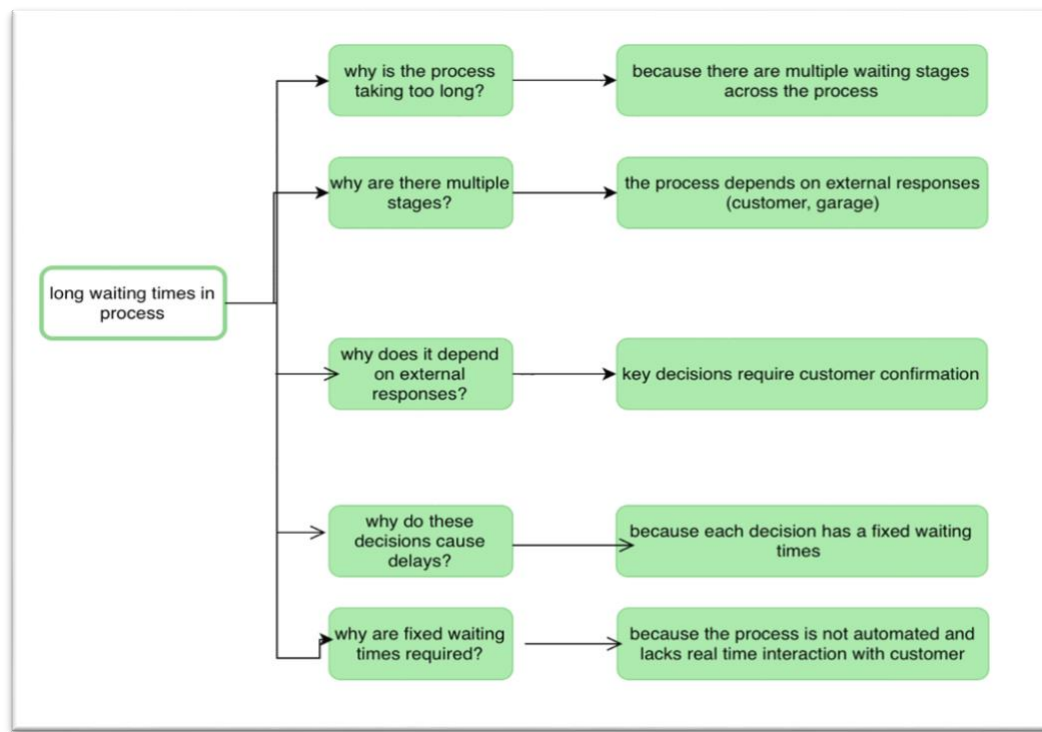


Figure 2 Root Analysis

Root cause: The process relies heavily on sequential, manual, and customer-dependent steps with fixed waiting periods, causing excessive delays and low efficiency.

7. Quantitative Analysis

7.1 Cycle Time and Processing Time

Task	Cycle Time	Processing Time
Register request	0.5 day	0.5 h
Verify warranty coverage	0.5 day	1 h
Handle invalid warranty	3 days	0.5 h
Perform initial inspection	1 day	2 h
Prepare quotation	1 day	2 h
Check availability	2 days	2 h
Book service appointment	1 day	1 h
Payment handling	7 days	0.5 h
Execute maintenance service	1 day	4 h

Table 7 Task Cycle Time and Processing Time

Assumption: The warranty decision gateway is estimated as 70% valid warranty and 30% invalid warranty. In addition, one working day is assumed to be equivalent to 8 working hours.

Overall Cycle Time = $0.5 + 0.5 + (3 * 0.30) + 1 + 1 + 2 + 1 + 7 + 1 = 15.9$ days

$15.9 * 8 = 127.2$ hours

Theoretical Cycle Time (TCT) = $0.5 + 1 + (0.5 * 0.30) + 2 + 2 + 2 + 1 + 0.5 + 4 = 13.15$ hours

Cycle Time Efficiency (CTE) = $13.15 \div 127.2 = 0.103 = 10.3\%$

This indicates **low process efficiency**, with most of the time spent on waiting and delays, highlighting clear opportunities for improvement.

7.2 Bottleneck identification

In the Request-to-Maintain process at Ford Service Center, bottleneck identification was conducted based on three sources.

1. The BPMN diagram was reviewed to locate timer events and decision gateways where the process stops and waits for an external response, such as the customer warranty decision and quotation approval steps.
2. The cycle time table was analyzed to identify tasks where the cycle time far exceeds the processing time for instance, the payment handling step has a cycle time of 7 days but a processing time of only 0.5 hours, meaning most of the time is spent waiting with no active work being done.
3. The overall Cycle Time Efficiency (CTE) of 10.3% confirms that 89.7% of the total process time is non-productive waiting time, which indicates the presence of significant bottlenecks that slow down the entire process.

7.3 Capacity analysis

Based on the process data:

- Annual requests = 500
- Working days = 250 days/year
- Working hours = 8 hours/day

Processing time per request:

- Initial inspection = 2 hours
- Quotation preparation = 2 hours
- Service booking = 1 hour
- Payment handling = 0.5 hour

Total processing time ≈ 5.5 hours per request.

Unit capacity per employee:

$$\mu = 1 / 5.5 = 0.18 \text{ requests/hour}$$

Assuming 3 employees:

$$\text{Total capacity } \mu = 3 \times 0.18 = 0.54 \text{ requests/hour}$$

Capacity per day:

$$0.54 \times 8 = 4.32 \text{ requests/day}$$

Capacity per year:

$$4.32 \times 250 = 1080 \text{ requests/year}$$

Interpretation:

The system capacity (1080 requests/year) exceeds the actual demand (500 requests/year), indicating that the process has sufficient capacity and is not constrained by resources.

7.4 Resource utilization:

Arrival rate:

$$\lambda = 500 / (250 \times 8) = 0.25 \text{ requests/hour}$$

Resource utilization:

$$\rho = \lambda / \mu = 0.25 / 0.54 \approx 0.46 \text{ (46\%)}$$

Interpretation:

The resource utilization is relatively low (46%), indicating that resources are not heavily loaded.

8. Process Redesign

We selected the Transactional Analytical Inward redesign approach because the main issues in the AS-IS process are caused by internal inefficiencies such as manual steps, delays, and lack of coordination. This approach focuses on improving internal operations through automation and better task structuring, enabling a faster, more efficient, and more reliable process.

8.1 Redesign Heuristics

Heuristic 1: Activity Automation (Time, Cost, Quality)

Manual activities such as request registration, warranty verification, and invoice handling are automated through the Ford System.

Impact: reduces processing time, lowers operational cost by minimizing manual effort, improves quality by reducing human errors, enhances customer experience through faster responses, and strengthens risk compliance by ensuring consistent rule-based processing.

Heuristic 2: Parallelism — "Put activities in parallel" (Time ,Cost)

In the AS-IS process, parts availability checking is performed only after quotation approval, causing delays. In the TO-BE design, the **Parts Availability Check is executed in parallel with Quotation Preparation** immediately after the initial inspection.

Impact: reduces delays by identifying parts availability earlier, lowers cost by avoiding late-stage rework, improves customer experience by preventing unexpected delays after quotation approval, and supports risk compliance by ensuring early validation of required resources before commitment.

Heuristic 3: Activity Composition (Time, Cost, Quality)

Separate tasks for booking a service bay and assigning a mechanic are combined into a single task: *Request Service Resources*.

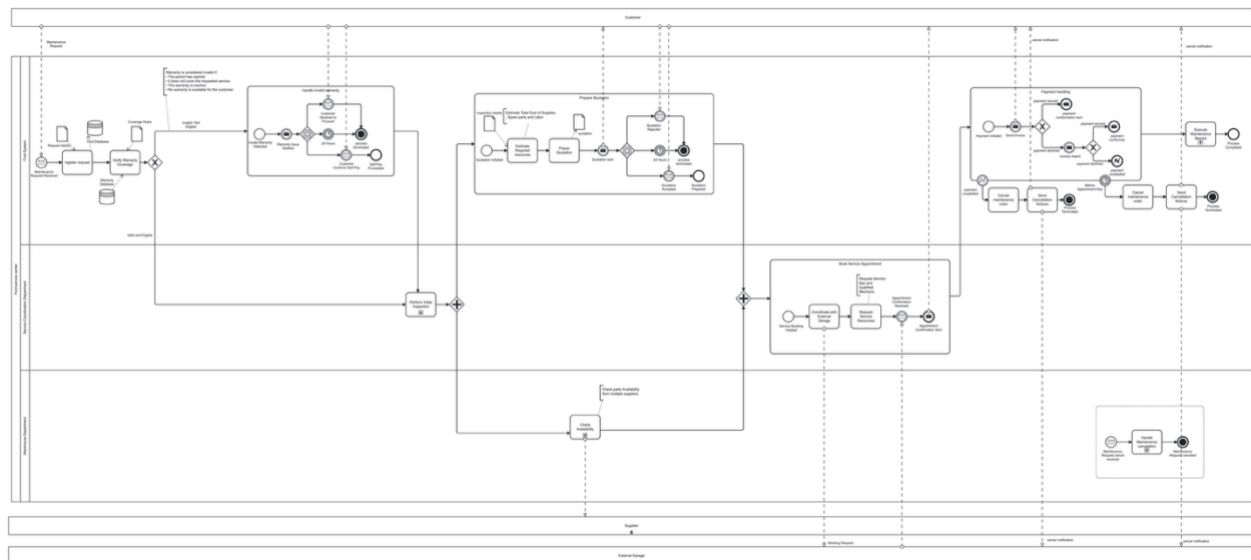
Impact: reduces time by eliminating unnecessary steps, lowers cost by reducing coordination effort, improves quality by minimizing scheduling conflicts, enhances customer experience through faster and clearer booking confirmation, and supports risk compliance by ensuring proper resource allocation in a single controlled step.

Heuristic 4: Case-Based Work (Time, Quality)

Time limits are introduced for key steps such as appointment confirmation (24 hours) and payment (1 week) to prevent delays.

Impact: reduces time by preventing process stagnation, improves quality through consistent case handling, enhances customer experience with predictable timelines, and strengthens risk compliance by enforcing deadlines and automatically closing incomplete cases.

8.2 TO-BE BPMN Model



To-Be Diagram3 Figure

For a clearer and more detailed view of the TO-BE diagram, please refer to the full-resolution version available via the link below.

<https://drive.google.com/file/d/1dHndyG2YSGINpTSRF0hgLBEJE0Aw9tgK/view?usp=drivesdk>

9. Implementation

The implementation presents the key automation tasks in the TO-BE Request-to-Maintain process. The proposed solutions replace manual activities with system-driven workflows through the Ford System, reducing errors and improving overall process efficiency.

1. Automated Maintenance Request Registration

When the customer submits a request, the Ford System automatically registers it, validates the data, and generates a unique case ID linked to the customer's vehicle profile. This eliminates manual data entry and ensures the request is instantly visible to all relevant departments.

2. Automated Warranty Verification

The Warranty Department Lane has been removed. The Ford System queries the Warranty Database and applies eligibility rules in real time without human involvement. If the warranty is invalid, the system notifies the customer immediately and provides the self-payment option.

3. Unified Service Resource Booking

The booking of the service bay and mechanic is merged into a single task: *Request Service Resources*. The Ford System sends one consolidated request to the External Garage with a 24-hour confirmation window.

4. Automated Invoice Generation and Payment Tracking

After appointment confirmation, the Ford System automatically generates and sends the invoice and activates a one-week payment timer. If payment is not completed within the timeframe, the system cancels the request and notifies all relevant parties.

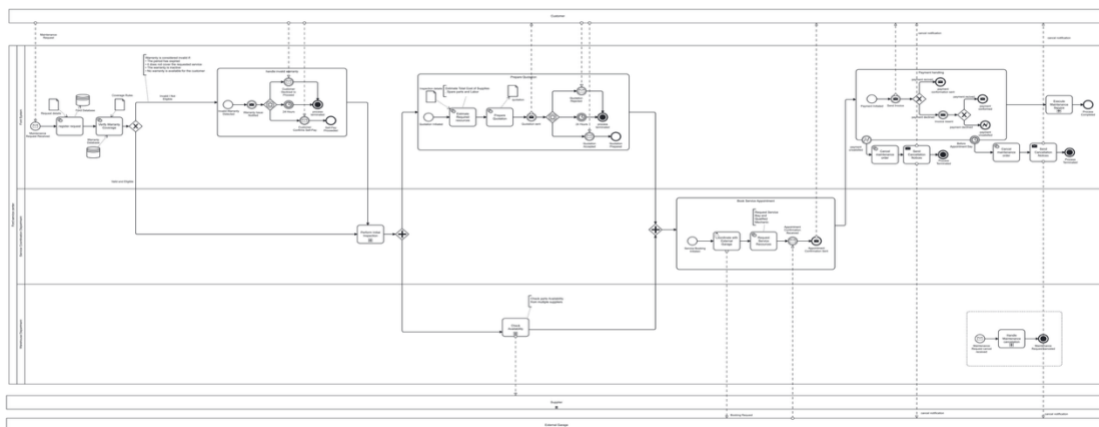


Figure 4 Implementation Diagram

For a clearer and more detailed view of the Implementation diagram, please refer to the full-resolution version available via the link below.

https://drive.google.com/file/d/1wJK-_34krJHVYRXsw88BNgMduEwvdctq/view?usp=drivesdk

10. Conclusion

The BPM redesign of the Request-to-Maintain process at Ford Service Center applies the **Transactional Analytical Inward** approach to address inefficiencies rooted in internal operations, such as long waiting times, manual processing steps, and lack of coordination between departments. Through process discovery, qualitative and quantitative analysis, the team identified excessive delays and high dependency on manual communication as the primary contributors to operational challenges.

The TO-BE model introduces four key improvements: automated warranty verification through the Ford System, parallel parts availability checking, unified service resource booking, and automated invoice generation with payment tracking. These changes collectively reduce waiting times, eliminate unnecessary manual steps, and enhance overall process efficiency and coordination.

Overall, the redesigned process strengthens operational performance and aligns with the Ford Service Center's goals for efficiency, service quality, and digital advancement.

11. Group Work Reflection

11.1 Team Meetings

Num	Date	Day	Mode	Discussion Topics
1	April 14	Tuesday	Online	Discussing the scenario, distributing roles, and defining process boundaries
2	April 15	Wednesday	Online	Building the As-Is BPMN diagram
3	April 16	Thursday	Online	Reviewing the As-Is diagram and refining sub-processes
4	April 19	Sunday	Online	Final review of the As-Is diagram and preparing the submission file
5	April 20	Monday	Online	Applying revisions based on instructor feedback received for Milestone 1
6	April 22	Wednesday	Online	Starting the Qualitative Analysis (Value-Added, Waste, and Stakeholder Analysis)
7	May 3	Sunday	Online	Completing the Quantitative Analysis and preparing the Milestone 2 submission file
8	May 4	Monday	On Campus	Applying revisions based on instructor feedback and beginning the To-Be diagram and implementation plan
9	May 9	Saturday	Online	Finalizing the To-Be BPMN diagram and writing the Redesign section
10	May 10	Sunday	Online	Final review of all deliverables and submission

Table 8 Meetings Times

11.2 Work Distribution

Task	Name
Process Context & Boundaries	All Team members
Process Profile & Stakeholders	All Team members
As-Is BPMN Diagram	All Team members
Process Discovery	All Team members
Value-Added Analysis	Rahaf , Wareef
Waste Analysis	Wareef , Reham
Stakeholder Analysis	Reham , Jouri
Issue Documentation	Reham , Jouri
Root Cause Analysis	Rahaf , Wareef
Quantitative Analysis	All Team members
To-Be BPMN Diagram	All Team members
Implementation Plan	All Team members

Table 9 Team Roles

12. References:

1. Titu, A.M., Grecu, D., Pop, A.B., & Şugar, I.R. (2025). Service Process Modeling in Practice: A Case Study in an Automotive Repair Service Provider. *Applied Sciences*, 15(8), 4171. <https://doi.org/10.3390/app15084171>
2. zur Muehlen, M., & Ho, D.T. (2008). Service Process Innovation: A Case Study of BPMN in Practice. *Proceedings of the 41st Hawaii International Conference on System Sciences*. <https://doi.org/10.1109/HICSS.2008.459>
3. Dumas, M., La Rosa, M., Mendling, J., & Reijers, H.A. (2018). *Fundamentals of Business Process Management* (2nd ed.). Springer.
4. Da Cruz, R. (n.d.). Auto Repair Process: 7 Basic Steps. *LinkedIn*. Retrieved April 13, 2026, from <https://www.linkedin.com/pulse/auto-repair-process-7-basic-steps-ricardo-da-cruz-1e>
5. BPMN Tutorial. (n.d.). YouTube. <https://youtu.be/5nrwocXtxCo?si=8dvYye1Aa9YP908A>
6. BPM Process Analysis. (n.d.). YouTube. <https://youtu.be/voNQBD Ae qtk?si=FEKbq lOtXxm6qdc m>
7. Alharbi, R., Alabdullatif, W., Alshamrani, R., & Alwagait, J. (2026, April 13). Direct observation of vehicle maintenance service workflow at Ford Service Center. King Saud University.